HDOC Day-49

Write a C program that reads a positive integer n. Pass n to a function named pivot_integer() using its address. Use pointers to pass the value between the calling function and pivot_integer(). The function pivot_integer() must:

- Find the pivot integer x between 1 and n such that the sum of all integers from 1 to x, inclusively, is equal to the sum of all integers from x to n, inclusively.
- Display the pivot integer x if it exists.
- Display -1 if no such pivot integer exists.

Example 1:

Input: n = 8

Output:

Pivot integer: 6 [Explanation: 1 + 2 + 3 + 4 + 5 + 6 = 6 + 7 + 8]

Example 2:

Input: n = 1

Output:

Pivot integer: 1 [Explanation: 1 = 1]

Example 3:

Input: n = 4

Output: -1 [Explanation: No pivot integer exists for n = 4]

Algorithm

1. Start.
2. Read a positive integer n.
3. Pass the address of n to the function pivot_integer().
4. Run a loop from x = 1 to n.
5. Calculate the sum from 1 to x and the sum from x+1 to n.

6. If the sum from 1 to x equals the sum from x+1 to n plus x, display x.
7. If no pivot integer is found, display -1.
8. Stop.

- **C Program:**

```
#include <stdio.h>

void pivot_integer(int *n)
{
   int x, left, right;

   for (x = 1; x <= *n; x++)
   {
      left = x * (x + 1) / 2;
      right = (*n) * (*n + 1) / 2 - left;

      if (left == right + x)
      {
         printf("Pivot integer: %d\n", x);
         return;
      }
   }

   printf("-1\n");
}

int main()
{
   int n;

   printf("Enter a positive integer: ");
   scanf("%d", &n);

   pivot_integer(&n);

   return 0;
}
```

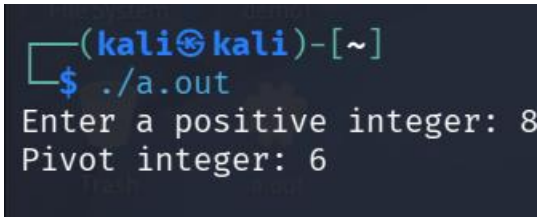
Test Case Table:

Test Case	Input n	Expected Output	Explanation
1	8	Pivot integer: 6	1+2+3+4+5+6=6+7+8+21
2	1	Pivot integer: 1	1=1
3	4	-1	No pivot integer exists
4	49	Pivot integer: 35	Sum from 1 to 35 equals sum from 36 to 49
5	2	-1	No pivot integer exists
6	6	-1	No pivot integer exists
7	3	-1	No pivot integer exists

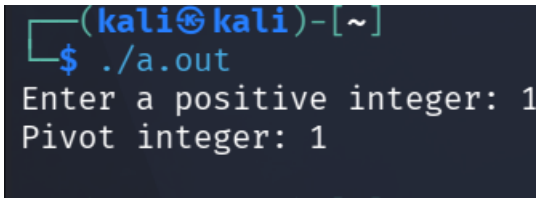
Compilation and Execution:

The following program has been compiled and executed successfully.

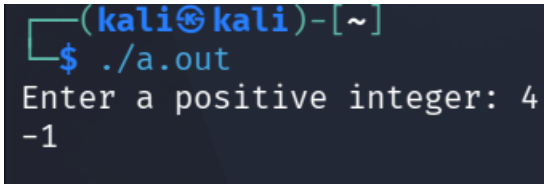
Test Case 1:



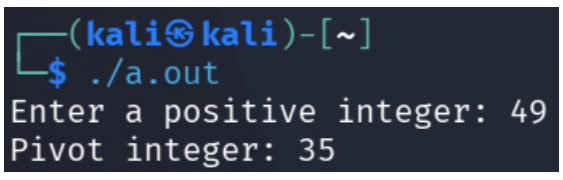
Test Case 2:



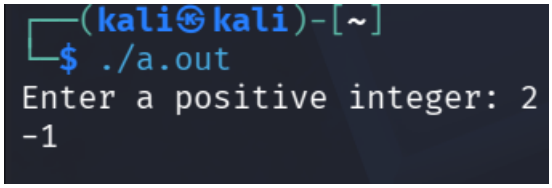
Test Case 3:



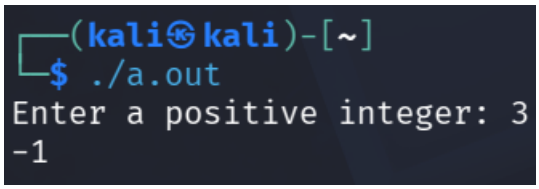
Test Case 4:



Test Case 5:



Test Case 6:



Test Case 7:

